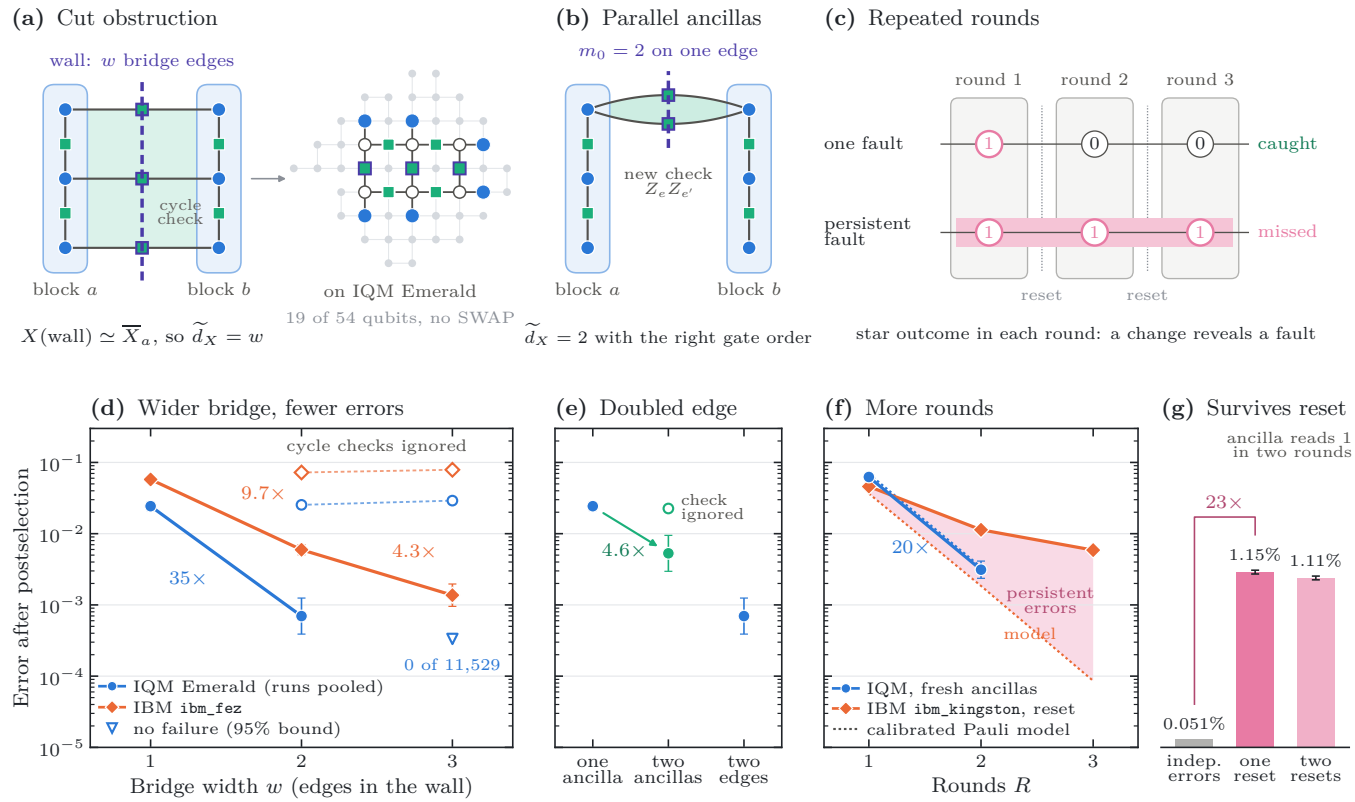

(a) Cut obstruction
wall: w bridge edges
cycle check
block a
block b
on IQM Emerald
19 of 54 qubits, no SWAP
X(wall) ≃ X̄a, so d̃X = w
(b) Parallel ancillas
m0 = 2 on one edge
new check
Ze Ze′
block a
block b
d̃X = 2 with the right gate order
(c) Repeated rounds
round 1
round 2
round 3
one fault
persistent fault
caught
missed
reset
reset
star outcome in each round: a change reveals a fault
(d) Wider bridge, fewer errors
cycle checks ignored
9.7×
4.3×
35×
0 of 11,529
IQM Emerald (runs pooled)
IBM ibm_fez
no failure (95% bound)
Error after postselection
Bridge width w (edges in the wall)
(e) Doubled edge
check ignored
4.6×
one ancilla
two ancillas
two edges
(f) More rounds
20×
persistent errors
model
IQM, fresh ancillas
IBM ibm_kingston, reset
calibrated Pauli model
Rounds R
(g) Survives reset
ancilla reads 1 in two rounds
23×
1.15%
1.11%
0.051%
indep. errors
one reset
two resets